

Workshop #2

Aeroservoelasticity of Fixed and Rotary Wing Aircraft

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1. **Pitch Flap Flutter using Quasi-steady aerodynamics:** Consider the Puma blade model. Take the first flap and first torsion modes. Assemble the aerodynamics matrices. Compute the eigenvalues and check the stability of the reference model.
2. **Pitch Flap Flutter using unsteady reference section aerodynamics:** Consider the Puma blade model. Take the first flap and first torsion mode. Assemble the aerodynamics matrices at the reduced frequency computed at $\frac{3}{4}$ of the chord. Compute the eigenvalues and check the stability of the reference model. Compute the variation of stability in percentage of change in damping ratio.
3. **Plot the stability map:** Consider the pitch link stiffness valued between 0 and the nominal value, and the position of the CG between 24% and 40% of the chord and plot the stability map, trying to identify the boundary between the stable and unstable configurations.
4. **Optional task 1 all modes.** Plot the stability map considering all elastic modes and not only the first flap and first torsion modes.
5. **Optional task 2 Exact unsteady.** Compute the stability map considering the exact reduced frequency for every section of the blade (hint: the K_{aer} and C_{aer} functions should be moved in the integrals to be performed along the span).